

Sidaard Gunasekaran

Associate Professor
Department of Mechanical and Aerospace Engineering
University of Dayton, OH 45469

gunasekarans1@udayton.edu
937-229-5345

OBJECTIVE

To advance the field of aeronautics by applying my research skills acumen and become an integral contributor in to an institution through prolific teaching methods and service that resonates in current millennial's learning methods.

EDUCATION/TRAINING

- Ph.D.** Aerospace Engineering, University of Dayton, August 2016.
Dissertation Title: *"Relationship Between the Wingtip Vortex, the Free Shear Layer and Aerodynamic Efficiency"*
- M.S.** Aerospace Engineering, University of Dayton, May 2012
Research: *"Analysis of Rapidly Pitching Wing using Superimposition of Rotational Circulation"*
- B.S.** Aerospace Engineering, SRM University, India, May 2011

PROFESSIONAL EXPERIENCE

Aug 2024 – Present: **Hans von Ohain Endowed Chair**, Department of Mechanical and Aerospace Engineering, University of Dayton, OH.

Aug 2023 – Present: **Associate Professor**, Department of Mechanical and Aerospace Engineering, University of Dayton, OH.

Aug 2016 – Present: **Director**, University of Dayton Low Speed Wind Tunnel, University of Dayton Water Tunnel, and University of Dayton Merlin Flight Simulator

Sept 2016 – 2023: **Assistant Professor**, Department of Mechanical and Aerospace Engineering, University of Dayton, Dayton, OH.

May 2015 – August 2016: **Graduate/Research Assistant, Department of Mechanical and Aerospace Engineering, University of Dayton, OH**

Taught undergraduate and graduate aerospace courses (Advance Aerodynamics, Engineering Materials and Introduction to Flight, Experimental aerodynamics). Assisted Dr. Aaron Altman in design and construction of experimental rig for various academic and research related experiments.

August 2013 – May 2015: **Graduate Lab Assistant, Department of Mechanical and Aerospace Engineering, University of Dayton, OH**

Facilitated and maintained Engineering Materials laboratory for students to perform experiments. Trained students to perform material testing in accordance with ASTM standards. Developed undergraduate composite laboratory module.

TEACHING EXPERIENCE

Taught the following courses at the University of Dayton

1. MEE 225 Introduction to Flight (6X)
2. MEE 312 Engineering Materials (2X)
3. MEE 312L Engineering Materials Lab (1X)
4. MEE 401/AEE 501 Fundamental Aerodynamics (10X)
5. MEE 425 Aircraft Design (5X)
6. MEE 498 Engineering Experimentation with Aero Specialization (1X)
7. AEE 555 Turbulence (1X)
8. AEE 553 Compressible Flow (3X)
9. AEE 590 Experimental Aerodynamics (5X)
10. AEE 590 Advanced Data Science and Machine Learning Applied to Aerodynamics (4X)

Professional Highlights

Member of KEEN's Team Crescendo <https://engineeringunleashed.com/card/10>

Selected to be one among 4 faculty across the network nationally to be in Team Crescendo bringing educators in institutions across the country a highly interactive mini-workshop experience that weaves together entrepreneurial mindset, joyful teaching, AI, and hands-on practice into one immersive package.

2025 EngineeringUnleashed Fellow <https://engineeringunleashed.com/fellows>

Design and Deployment of Case Studies with iFLY for Core Technical Courses

Speaker on the KEEN National Conference keynote on Designing Case Studies with iFLY Skydiving Wind Tunnel for Core Technical Courses <https://www.youtube.com/watch?v=xQYK3G58xOk&t=1193s> (My entry: 4:55 in the video)

2023 KEEN TOP NATIONAL Rising Star Award

"Story: 2023 KEEN Rising Star: Sid G | Engineering Unleashed." Engineeringunleashed.com, [Story: 2023 KEEN Rising Star: Sid G | Engineering Unleashed](#). Accessed 27 Jan. 2024.

Rock Star. Rock Star : University of Dayton, Ohio. (n.d.).

https://udayton.edu/blogs/president/2024/03/rock_star.php . Accessed 30 Mar. 2024.

Associate Professor Receives Kern Entrepreneurial Engineering Network Rising Star Award | University of Dayton, Ohio [Associate Professor Receives Kern Entrepreneurial Engineering Network Rising Star Award : University of Dayton, Ohio](#). Accessed 26 Oct 2024.

[University of Dayton School of Engineering Impact Report - Spring 2024](#)

Project Based Learning and Inquiry Learning in Aerodynamics using Passion Projects:

Garrow, K. *New classroom approach brings passion to the forefront of learning: University of Dayton, Ohio*. Retrieved June 18, 2022, from <https://bit.ly/PassionProjectMEE401>

Design, Build, Test (Learning by Design) in Introduction to Flight:

Updyke, K. (June 10, 2021). *When passion takes flight*. University of Dayton, Ohio. Retrieved June 18, 2022, from <https://bit.ly/WhenPassionTakesFlight>

Problem and Project Based Learning in Aircraft Design using Flight Simulator:

Merlin Flight Simulator Lab. Curriculum : University of Dayton, Ohio. Retrieved June 18, 2022, from <https://bit.ly/MerlinFlightSimulator>

Use of LTC Studio:

Curran, A. *A Look Inside the LTC studio with Dr. Sidaard Gunasekaran*. Listen Notes. Retrieved June 18, 2022, from <https://bit.ly/TeachinginLTCStudio>

Integration of Entrepreneurial Minded Learning (EML):

Our Competitive Edge : University of Dayton, Ohio. Retrieved June 18, 2022, from <https://bit.ly/EMLatUD>

Use of IDA Pedagogy in MEE 225 Intro to Flight Featured in Aerospace America:

December 2017 AIAA bulletin. Aerospace America. (2018, June 8). Retrieved June 18, 2022, from <https://bit.ly/IDAinMEE225>

Measuring Mindset using Portfolios

KEEN Network (2019, March 14). *How do you measure a mindset? | Sid Gunasekaran*. YouTube. Retrieved June 18, 2022, from <https://bit.ly/MeasuringMindset>

VISION Award:

Zandvoort, A. (2019) *Faculty and staff recognized with 2018 vision awards*. Faculty and Staff Recognized with 2018 Vision Awards : University of Dayton, Ohio. Retrieved June 18, 2022, <https://bit.ly/VisionAwardSid>

THESIS MENTORSHIP

Undergraduate Honors Thesis Mentorship (Clare-Booth Luce Fellowship)

1. Loughnane, Faith A., "Effect of Airfoil-Preserved Undulations on Wing Performance and Wingtip Vortex" (2020). *Honors Theses*. 300. https://ecommons.udayton.edu/uhp_theses/300
2. Culpepper, Grace G., "Propeller and Propeller-In-Wing Thrust Vectoring" (2021). *Honors Theses*. 311. https://ecommons.udayton.edu/uhp_theses/311
3. Foster, Michael R., "Investigating Tandem Propeller Performance in Multirotor Aircraft" (2025). *Honors Theses*. 471. https://ecommons.udayton.edu/uhp_theses/471

Master's Thesis Mentorship

1. Goodman, Steven. "Correlations Between Near Wake Velocity Fluctuations and Aerodynamic Efficiency for the SD7003 Airfoil." Master's thesis, University of Dayton, 2018. **(Primary Advisor)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1524137507116155
2. Pleiman, Brock. "Calibration of a Flow Angularity Probe with a Real-Time Pressure Sensor." Master's thesis, University of Dayton, 2019. **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1568119292467936

3. Sholtis, Paul. *"Characterization of Upstream Effects Due to High Blockage in the AFRL Vertical Wind Tunnel"*, Master's Thesis, University of Dayton, 2019. **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton155594746187138
4. Cai, Jielong. *"Changes in Propeller Performance Due to Ground and Partial Ground Proximity."* Master's thesis, University of Dayton, 2020. **(Primary Advisor)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1588164898961792
5. Insana, Eric. *"Low Reynolds Number Experimental Aerodynamic Verification of Scaled and LEWICE Simulated Ice Accretions in SLD Conditions."* Master's thesis, University of Dayton, 2020. **(Primary Advisor)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1597082782224451
6. Barnhart, Samuel. *"Design and Development of a Coherent Detection Rayleigh Doppler Lidar System for Use as an Alternative Velocimetry Technique in Wind Tunnels."* Master's thesis, University of Dayton, 2020. **(Primary Advisor)** http://rave.ohiolink.edu/etdc/view?acc_num=dayton1595276402371536
7. Deslich, Joshua. *"Effects of a Bell-Shaped Lift Distribution on an Oblique Flying Wing and its Impact on Aerodynamic Performance."* Master's thesis, University of Dayton, 2020. **(Primary Advisor)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1590560243607956
8. Lickenbrock, Madeline. *"Multi-fidelity, Multidisciplinary Design Analysis and Optimization of the Efficient Supersonic Air Vehicle."* Master's thesis, University of Dayton, 2020. **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1608156251624819
9. Moidel, Benjamin. *"Development and Validation of Single-Engine General Aviation Aircraft Models within Merlin 521 Motion-Based Flight Simulators."* Master's thesis, University of Dayton, 2021. **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1619738645033377
10. Mongin, Michael. *"Closed Loop Control of Lift Coefficient at High Amplitudes and at Different Reduced Frequencies."* Master's thesis, University of Dayton, 2021. **(Primary Advisor)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1639590733428631
11. Killian, Andrew. *"Periodic Vortical Gust Encounter and Mitigation using Closed Loop Control"*. Master's thesis, University of Dayton, 2023. **(Primary Advisor)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1682515173209394
12. Tierney, Ian. *"Measuring Agricultural Spray Droplet Distribution in Propeller Wake: A Cautionary Tale"*. Master's thesis, University of Dayton, 2023. **(Primary Advisor)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1683046679523457
13. Settle, Michael. *"The Effects of Deployable Surface Topography Using Liquid Crystal Elastomers on Cylindrical Bodies in Flow"*. Master's Thesis, University of Dayton, 2023. **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1683048659500655
14. Jestus, Nevin. *"Aerodynamic Characterization of Multiple Wing-Wing Interactions for Distributed Lift Applications."* Master's thesis, University of Dayton, 2023. **(Primary Advisor)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1690882021109845

15. Natalie Douglas. “Granular Media Motion in a Rotary Kiln: Design, Testing, and Analysis Combined with Coupled Radiative Exchange Techniques in a Discrete Element Method Tool for Concentrating Solar Power Application.” Master’s thesis, University of Dayton, 2023 **(Committee Member)**

16. Lucas Duncan. “Powered Wing Response to Streamwise Gust Encounters.” Master’s thesis, University of Dayton, 2024 http://rave.ohiolink.edu/etdc/view?acc_num=dayton1714558305349343 **(Primary Advisor)**

17. Sushil Janardhana, “Unfiltered Rayleigh Scattering and Synthetic Schlieren for Thermodynamic Characterization of Heated Binary Jets”, University of Dayton 2025 **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1754425006728733

Ph.D. Dissertation Mentorship:

1. Quershi, Saad. “Predicting Aerially Delivered Retardant Ground Deposit Concentrations and Spatial Distributions using Statistical and Algebraic Modeling with Influence from Experimental Techniques”. Ph.D. Dissertation, University of Dayton, 2022 **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1650795309937635

2. Donovan, Molly. “Impact of Free-stream Turbulence Intensity on the End wall Region of Low-Pressure Turbine Blades”. Ph.D. Dissertation, University of Dayton, 2023 **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1681823544653736

3. Osama Alsattam, “Noise Robust Particle Event Velocimetry with A Kalman Filter-Based Tracking”. Ph.D. Dissertation, University of Dayton, 2024 **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1714134260767049

4. Katie Opacich, “Investigation of Ignition Kernel to Kernel Interactions by Utilizing Imaging Diagnostics around/within a Small Flow Tunnel”. Ph.D. Dissertation, University of Dayton, 2024 **(Committee Member)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1714650915201062

5. Jielong Cai, “Changes in Propeller Performance due to Rotor and Ceiling Proximity in Forward Flight”. Ph.D. Dissertation, University of Dayton, 2024 **(Primary Advisor)**
http://rave.ohiolink.edu/etdc/view?acc_num=dayton1714143735792499

Current Undergraduate Students:

1. Jessica DeMoor, ‘Flight Test Validation of Tandem Propeller Performance with Vertical Offset’
2. Samuel Parlett, ‘Experimental Investigation of a Variable Pitch Propeller at Various Incidence Angles in Forward Flight’
3. Brock Greenwood, ‘Characterization of Agricultural Nozzle Spray Geometry in Propeller Wake’
4. Grace Schreyer, ‘Exploration of the Presence of Nearly Two-Dimensional Wake Structure of Non-Elliptical Lift Distributions’.
5. Grace Selm, ‘Design and Testing of Morphing Wing with Spanwise Twist’
6. Megan Kelly, ‘Flight Test Validation of Tandem Propeller Performance with Vertical Offset’
7. Megan Bender, ‘Characterization of Agricultural Nozzle Spray Geometry in Propeller Wake’
8. Tyler Worthington, ‘Evaluation of Wing-Embedded Propulsor Systems in Maritime Ground Effect’
9. Hai Long Le, ‘Characterization of Agricultural Nozzle Spray Geometry in Propeller Wake’
10. Nolan Elo, ‘Event-based Characterization of focused-ultrasound-induced acoustic streaming’

11. Autumn Randall, 'Flight Test Validation of Tandem Propeller Performance with Vertical Offset'
12. David Agyei, 'Event-based Characterization of focused-ultrasound-induced acoustic streaming'

Current Masters students:

1. Julian Pabon, 'Experimental Investigation of Dynamic Pitching Effects on a Delta Wing with Blown Jet'
2. Joe Kastner, 'Morphological Transitions in Droplet Breakup Under Shock-Induced Flows'

Current Ph.D students:

1. Josh Deslich, "Multi-Fidelity and Multi-Disciplinary Design Optimization of a Flying Wing" (**Committee Member**)
2. Charles C.B Cain, "Tailoring Lift Distribution of Wings through Non-Elliptical Spanwise Twist and Active Flow Control" (**Primary Advisor**)

RESEARCH PUBLICATIONS

Magazine Articles

1. **Gunasekaran, Sidaard** and Doug Melton. "Getting EML Off the Ground: Why Do Jets Fly So High?". KEEN Magazine, p 08. 2017 (<http://online.fliphtml5.com/zyet/ykie/#p=8>)

Peer-Review Journal Articles

Published:

1. **Gunasekaran, Sidaard**, and Daniel Curry. "On the Wake Properties of Segmented Trailing Edge Extensions." *Aerospace* 5, no. 3 (2018): 89. <https://doi.org/10.3390/aerospace5030089>
2. **Gunasekaran, Sidaard**, and Tim Gerham. "Effect of Chordwise Slots on Aerodynamic Efficiency and Wingtip Vortex." *AIAA Journal* 56, no. 12 (2018): 4752-4767. <https://doi.org/10.2514/1.J057073>
3. **Gunasekaran, Sidaard**, and Grant Ross. "Effect of Piezo-Embedded Inverted Flag in Free Shear Layer Wake." *Aerospace* 6, no. 3 (2019): 33. <https://doi.org/10.3390/aerospace6030033>
4. Cai, Jielong, **Sidaard Gunasekaran**, Michael OL, and Anwar Ahmed. "Propeller Ground and Ceiling Effect Parametric Data." *Journal of Aircraft* (2020): 1-4. <https://doi.org/10.2514/1.C035839>
5. Loughnane, Faith, Michael Mongin, **Sidaard Gunasekaran**, "Effect of Airfoil-Preserved Undulation Placement on Wing Performance and Wingtip Vortex", *AIAA Journal* <https://doi.org/10.2514/1.J059517>
6. **Gunasekaran, Sidaard**, and Aaron Altman. "Far Wake and Its Relation to Aerodynamic Efficiency." *Energies* 14, no. 12 (2021): 3641. <https://doi.org/10.3390/en14123641>
7. **Gunasekaran, Sidaard**, Madison Peyton, and Neal Novotny. 2022. "Aerodynamic Interactions of Wind Lenses at Close Proximities" *Energies* 15, no. 13: 4622. <https://doi.org/10.3390/en15134622>
8. Cai, Jielong, **Sidaard Gunasekaran**, and Michael OL. "Effect of Partial Ground and Partial Ceiling on Propeller Performance." *Journal of Aircraft* (2022): 1-14. <https://doi.org/10.2514/1.C036974>
9. Cai, Jielong, and **Sidaard Gunasekaran**. "Propeller Ground Effect in Forward Flight." *AIAA Journal* 61, no. 10 (2023): 4378-4390. <https://doi.org/10.2514/1.J062984>
10. Cai, Jielong, **Sidaard Gunasekaran**, and Michael OL. "Effect of Partial Ground and Partial Ceiling on Propeller Performance." *Journal of Aircraft* 60, no. 3 (2023): 648-661. <https://doi.org/10.2514/1.C036974>

11. Cai, Jielong, and **Sidaard Gunasekaran**. "Frequency Response of RC Propellers to Streamwise Gusts in Forward Flight." *Wind* 3, no. 2 (2023): 253-272. <https://doi.org/10.3390/wind3020015>
12. AlSattam, Osama, Michael Mongin, Mitchell Grose, **Sidaard Gunasekaran**, and Keigo Hirakawa. "KF-PEV: a causal Kalman filter-based particle event velocimetry." *Experiments in Fluids* 65, no. 9 (2024): 141. <https://doi.org/10.1007/s00348-024-03877-y>
13. Cai, Jielong, and **Sidaard Gunasekaran**. "Propeller Ceiling Effect in Forward Flight." *Journal of Aircraft* (2024): 1-17. <https://doi.org/10.2514/1.C037779>
14. **Gunasekaran, Sidaard**, Michael Mongin, Osama A. AlSattam, and Keigo Hirakawa. "Comparison of Event-Based Algorithms for Experimental Two-Dimensional, Two-Component Velocimetry." *Journal of Aircraft* (2025): 1-15. <https://doi.org/10.2514/1.C038412>

Peer-Reviewed Conference Articles

1. Lego, Zachary, Aaron Altman, and **Sidaard Gunasekaran**. "Experimental and Computational Analysis of High Angle of Attack Perching Maneuvers." *30th AIAA Applied Aerodynamics Conference*, AIAA 2012-2666. <https://doi.org/10.2514/6.2012-2666>
2. Altman, Aaron, and **Sidaard Gunasekaran**. "Identification of Aircraft by their Unique Turbulent Wake Signature: Progress with Experimental Validation." In *51st AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition*, AIAA 2012-1120. <https://doi.org/10.2514/6.2013-1120>
3. **Gunasekaran, Sidaard**, and Aaron Altman. "Wing Performance Insight from the Self-Preserved Turbulent Wake." *52nd Aerospace Sciences Meeting*, AIAA 2014-0247. <https://doi.org/10.2514/6.2014-0247>
4. **Gunasekaran, Sidaard**, Aaron Altman, and Kenneth O. Granlund. "Trends in Early Vortex Formation on a Wall-to-wall Plate in Pure Plunge." *53rd AIAA Aerospace Sciences Meeting*, AIAA 2015-1753. <https://doi.org/10.2514/6.2015-1753>
5. **Gunasekaran, Sidaard**, Aaron Altman, and Michael V. Ol. "Errors in Off-axis Loading of Off-the-shelf 6-Component Force Transducers: A Cautionary Tale." *53rd AIAA Aerospace Sciences Meeting*, AIAA 2015-1562. <https://doi.org/10.2514/6.2015-1562>
6. **Gunasekaran, Sidaard**, Gretchen Berkemeier[†], and Margaret Pinnell. "Design of a New Lab on Three Point Bend Test on Fiberglass Composite", ASEE North Central Section Conference. 2015.
7. **Gunasekaran, Sidaard**, and Aaron Altman. "Is There a Relationship Between the Wingtip Vortex and the Shear Layer?." *54th AIAA Aerospace Sciences Meeting*, AIAA 2016-1068. <https://doi.org/10.2514/6.2016-1068>
8. **Gunasekaran, Sidaard**, and Aaron Altman. "Better Insight into the Wingtip Vortex-Free Shear Layer Interaction." *55th AIAA Aerospace Sciences Meeting*, AIAA 2017-1705. <https://doi.org/10.2514/6.2017-1705>
9. **Gunasekaran, Sidaard**. "Integrated Teaching Model in Graduate Aerospace Classes: A Trial With Compressible Flow Aerodynamics." *ASEE Annual Conference and Exposition*. 2017. <https://doi.org/10.18260/1-2--28546>
10. **Gunasekaran, Sidaard**, and Aaron Altman. "Insight into Wingtip Vortex-Wing Free Shear Layer Interaction." *10th International Symposium on Turbulence and Shear Flow Phenomenon (TSFP10)*, 2017.
11. **Gunasekaran, Sidaard**, and Tim Gerham. "Effect of Chordwise Slots on Aerodynamic Efficiency." *2018 AIAA Aerospace Sciences Meeting*, p. AIAA 2018-0342. <https://doi.org/10.2514/6.2018-0342>

12. **Gunasekaran, Sidaard** and Daniel Curry. "Effect of Segmented Trailing Edge Extensions on Aerodynamic Efficiency." *2018 AIAA Aerospace Sciences Meeting*, AIAA 2018-0343. <https://doi.org/10.2514/6.2018-0343>
13. **Gunasekaran, Sidaard** and Nathan Thomas. "Affecting Aerodynamic Efficiency by Influencing Wing Surface-Flow Direction." *2018 AIAA Aerospace Sciences Meeting*, AIAA 2018-0344. <https://doi.org/10.2514/6.2018-0344>
14. **Gunasekaran, Sidaard**. "Integrated Teaching Model: A Follow Up with Fundamental Aerodynamics." *ASEE Annual Conference and Exposition*. 2018. <https://peer.asee.org/30677>
15. **Gunasekaran, Sidaard**, Grant V. Ross, and Daniel Curry. "Effect of Flexible Inverted PVDF in Free Shear Layer Wake." In *AIAA Scitech 2019 Forum*, p. 1338. 2019. <https://doi.org/10.2514/6.2019-1338>
16. Goodman, Steven, **Sidaard Gunasekaran**, Aaron Altman, and Albert Medina. "On the Near Wake Turbulent Flow Properties of the SD7003 Airfoil." In *AIAA Scitech 2019 Forum*, p. 0073. 2019. <https://doi.org/10.2514/6.2019-0073>
17. **Gunasekaran, Sidaard**. "Using Ideation, Disruption, Aha (IDA) Pedagogy in Introduction to Flight." *ASEE Annual Conference and Exposition*. 2019. <https://peer.asee.org/25203>
18. Palmer, Asa E., and **Sidaard Gunasekaran**. "Effect of Curved Boundary Layer Fences on Aerodynamic Efficiency." In *AIAA Scitech 2019 Forum*, p. 1614. 2019. <https://doi.org/10.2514/6.2019-1614>
19. Cai, Jielong, **Sidaard Gunasekaran**, Anwar Ahmed, and Michael V. Ol. "Changes In Propeller Performance Due to Ground Proximity." In *AIAA Scitech 2019 Forum*, p. 1097. 2019. <https://doi.org/10.2514/6.2019-1097>
20. Deslich, Joshua, and Sidaard Gunasekaran. "Effect of Slotted Winglet on the Wingtip Vortex." In *AIAA Aviation 2019 Forum*, p. 3431. 2019. <https://doi.org/10.2514/6.2019-3431>
21. Loughnane, Faith A., Rachael Supina, Michael P. Mongin, and **Sidaard Gunasekaran**. "Effect of Airfoil-Preserved Undulations on Wing Performance." In *AIAA Scitech 2020 Forum*, p. 1784. 2020. <https://doi.org/10.2514/6.2020-1784>
22. Cook, Thomas, **Sidaard Gunasekaran**, Michael V. Ol, and Michael P. Mongin. "Frequency Response of a Shuttered Open Jet Wind Tunnel." In *AIAA Scitech 2020 Forum*, p. 1760. 2020. <https://doi.org/10.2514/6.2020-1760>
23. Novotny, Neal, and **Sidaard Gunasekaran**. "Performance and Proximity Investigations on Small Scale Lensed Turbines." In *AIAA Scitech 2020 Forum*, p. 1242. 2020. <https://doi.org/10.2514/6.2020-1242>
24. Barnhart, Samuel A., and **Sidaard Gunasekaran**. "Investigation of Doppler Lidar for Velocity Measurements in Wind Tunnels." In *AIAA Scitech 2020 Forum*, p. 1021. 2020. <https://doi.org/10.2514/6.2020-1021>
25. Gazella, Matthew R., Jonathon Hill, Scott Chriss, Austin J. Abel, and **Sidaard Gunasekaran**. "Design and Analysis of an Additive Manufactured Supersonic Wind Tunnel." In *AIAA Scitech 2020 Forum*, p. 1354. 2020. <https://doi.org/10.2514/6.2020-1354>
26. Deslich, Josh, **Sidaard Gunasekaran**, Peter Flick, and Douglas Szczublewski. "Effects of spanloading and slew angle on an Oblique Flying Wing." In *AIAA Scitech 2020 Forum*, p. 0532. 2020. <https://doi.org/10.2514/6.2020-0532>
27. Cai, Jielong, **Sidaard Gunasekaran**, Anwar Ahmed, and Michael V. Ol. "Propeller Partial Ground Effect." In *AIAA Scitech 2020 Forum*, p. 1028. 2020. <https://doi.org/10.2514/6.2020-1028>

28. Mongin, Michael P., and **Sidaard Gunasekaran**. "Lift Regulation Using Closed-Loop Feedback and Control." In AIAA Scitech 2021 Forum, p. 2000. 2021. <https://doi.org/10.2514/6.2021-2000>
29. Culpepper, Grace G., Jielong Cai, and **Sidaard Gunasekaran**. "Propeller and Propeller-in-Wing Thrust Vectoring." In AIAA Scitech 2021 Forum, p. 0615. 2021. <https://doi.org/10.2514/6.2021-0615>
30. Cai, Jielong, **Sidaard Gunasekaran**, and Michael V. Ol. "Propeller Partial Ground and Ceiling Effect Prediction." In AIAA Scitech 2021 Forum, p. 0614. 2021. <https://doi.org/10.2514/6.2021-0614>
31. Sarnhart, Samuel A., Barath Narayanan, and **Sidaard Gunasekaran**. "Blown Wing Aerodynamic Coefficient Predictions Using Traditional Machine Learning and Data Science Approaches." In AIAA Scitech 2021 Forum, p. 0616. 2021. <https://doi.org/10.2514/6.2021-0616>
32. **Gunasekaran, Sidaard**, and Rachel Sharp. "Airfoil Near Wake Turbulent Properties at Maximum Aerodynamic Efficiency Condition." In AIAA Scitech 2021 Forum, p. 1848. 2021. <https://doi.org/10.2514/6.2021-1848>
33. Loughnane, Faith A., Michael P. Mongin, and **Sidaard Gunasekaran**. "Effect of Airfoil-Preserved Undulations on Free Shear Layer." In AIAA Scitech 2021 Forum, p. 0847. 2021. <https://doi.org/10.2514/6.2021-0847>
34. Ivarson, Joseph T., Lars Maneck, Barath Narayanan Narayanan, and **Sidaard Gunasekaran**. "Deep Learning Algorithm for Atomization Characterization using Shadowgraph Images." In AIAA SCITECH 2022 Forum, p. 0188. 2022. <https://doi.org/10.2514/6.2022-0188>
35. Donovan, Molly H., Markus P. Rumpfkeil, **Sidaard Gunasekaran**, and Christopher R. Marks. "Analysis of High-lift Low Pressure Turbine Endwall Vortices Using Modal Decomposition Methods." In AIAA SCITECH 2022 Forum, p. 0477. 2022. <https://doi.org/10.2514/6.2022-0477>
36. Cai, Jielong, **Sidaard Gunasekaran**, and Michael V. Ol. "Sinusoidal Gust Response of RC Propellers at Different Incidence Angles." In AIAA SCITECH 2022 Forum, p. 1536. 2022. <https://doi.org/10.2514/6.2022-1536>
37. Peyton, Madison N., and **Sidaard Gunasekaran**. "Investigation into Wake Interactions of Wind Lenses at Close Proximities." In AIAA SCITECH 2022 Forum, p. 1789. 2022. <https://doi.org/10.2514/6.2022-1789>
38. Culpepper, Grace G., Jielong Cai, and **Sidaard Gunasekaran**. "Effect of Forward Propeller Tilt and Inlet Shape in Propeller-in-Wing Thrust Vectoring." In AIAA SCITECH 2022 Forum, p. 1539. 2022. <https://doi.org/10.2514/6.2022-1539>
39. Hung, Lucas K., Jielong Cai, Neal Novotny, and **Sidaard Gunasekaran**. "Development of a Multi-Purpose Flap and Spoiler Mechanism for High Endurance Unmanned Aerial Vehicles." In AIAA SCITECH 2022 Forum, p. 1027. 2022. <https://doi.org/10.2514/6.2022-1027>
40. Insana, Eric, and **Sidaard Gunasekaran**. "Low Reynolds Number Experimental Aerodynamic Verification of Scaled and LEWICE Simulated Ice Accretions in SLD Conditions." In AIAA SCITECH 2022 Forum, p. 2025. 2022. <https://doi.org/10.2514/6.2022-2025>
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43. Cai, Jielong., **Sidaard Gunasekaran**, "Propeller Ground and Ceiling Effect in Forward Flight", Vertical Flight Society, Forum 78, Dallas, TX, USA, 2022. [SKU # 78-2022-0072-Cai](#)
44. Killian, Andrew, **Sidaard Gunasekaran**, Michael P. Mongin, and Albert Medina. "Periodic Vortical Gust Encounter and Mitigation Using Closed Loop Control." In *AIAA SCITECH 2023 Forum*, p. 2477. 2023. <https://doi.org/10.2514/6.2023-2477>
45. Mongin, Michael P., Albert Medina, Andrew Killian, and **Sidaard Gunasekaran**. "High Amplitude Lift Tracking Using Closed-Loop Feedback and Control; A Flow Analysis." In *AIAA SCITECH 2023 Forum*, p. 2476. 2023. <https://doi.org/10.2514/6.2023-2476>
46. Tierney, Ian M., **Sidaard Gunasekaran**, Kyle Butz, Timothy Anderson, and C. Taber Wanstall. "Measuring Agricultural Spray Droplet Distirbutions in Propeller Wake: A Cautionary Tale." In *AIAA SCITECH 2023 Forum*, p. 0090. 2023. <https://doi.org/10.2514/6.2023-0090>
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49. Gogidze, George J., **Sidaard Gunasekaran**, and Jielong Cai. "Effect of Propeller Incidence Angle on Wing Embedded Propeller Configuration in Forward Flight." In *AIAA SCITECH 2023 Forum*, p. 2284. 2023. <https://doi.org/10.2514/6.2023-2284>
50. Mongin, Michael P., Aaron Altman, and **Sidaard Gunasekaran**. "Extended High Lift Characteristics of Distributed Lift Configurations." In *AIAA SCITECH 2023 Forum*, p. 0031. 2023. <https://doi.org/10.2514/6.2023-0031>
51. Cai, Jielong, and **Sidaard Gunasekaran**. "Sinusoidal Gust Response of RC Propellers in Tandem Configuration" In VFS Forum 79, <https://doi.org/10.4050/F-0079-2023-17981>
52. Pabon, Julian A., Xinyu Gao, Jielong Cai, and **Sidaard Gunasekaran**. "Experimental Investigation of a Novel Morphing Wing Design." In *AIAA 2023 Region III Student Conference*, Dayton, OH <https://doi.org/10.2514/6.2023-71704>
53. Duncan, Lucas, Jielong Cai, and **Sidaard Gunasekaran**. "Powered Wing's Response to Streamwise Gust Encounters." In *AIAA 2023 Region III Student Conference*, Dayton, OH <https://doi.org/10.2514/6.2023-71287>
54. Jestus, Nevin, **Sidaard Gunasekaran**, Michael P. Mongin, and Aaron Altman. "Aerodynamic Interactions among Three Identical Wings in Close Proximity." In *AIAA 2023 Region III Student Conference*, Dayton, OH <https://doi.org/10.2514/6.2023-71488>
55. Porterfield, Andrew, Andrew Killian, **Sidaard Gunasekaran**, and Michael Mongin. "Generation and Characterization of Discrete Vortical Gust". In *AIAA 2023 Region III Student Conference*, Dayton, OH <https://doi.org/10.2514/6.2023-71873>
56. Cai, Jielong, **Sidaard Gunasekaran**, and Michael Ol. "Investigation of Positively Staggered Vertically Offset Propellers in Tandem." In *AIAA SCITECH 2024 Forum*, p. 1510. 2024. <https://doi.org/10.2514/6.2024-1510>

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58. Vincent, Timothy, **Sidaard Gunasekaran**, Michael Mongin, Alberto Medina, Alexander M. Pankonien, and Philip Buskohl. "Development of an Experimental Testbed to Study Cavity Flow as a Processing Element for Flow Disturbances." In *AIAA SCITECH 2024 Forum*, p. 1732. 2024. <https://doi.org/10.2514/6.2024-1732>
59. Kulig, Jacob, Jielong Cai, **Sidaard Gunasekaran**, and Carson L. Running. "Back-Imaging of Pressure-Sensitive Paint to Determine Close Proximity Ground Effects of Propellers." In *AIAA SCITECH 2024 Forum*, p. 0878. 2024. <https://doi.org/10.2514/6.2024-0878>
60. Pabon, Julian A., Xinyu Gao, Jielong Cai, and **Sidaard Gunasekaran**. "Experimental Investigation of a Novel Morphing Wing Design." In *AIAA SCITECH 2024 Forum*, p. 1349. 2024. <https://doi.org/10.2514/6.2024-1349>
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63. Cai, Jielong, Caiden Guzman, Emma Schutter, and **Sidaard Gunasekaran**. "On the Linear Superposition of Wing and Propeller Performance in a Wing Embedded Propeller System." In *AIAA SCITECH 2024 Forum*, p. 1511. 2024. <https://doi.org/10.2514/6.2024-1511>
64. AlSattam, Osama A., Michael P. Mongin, Andrew Killian, **Sidaard Gunasekaran**, and Keigo Hirakawa. "Toward Event-Based Noise-Robust High Density Particle Velocimetry." In *AIAA SCITECH 2024 Forum*, p. 2663. 2024. <https://doi.org/10.2514/6.2024-2663>
65. **Gunasekaran, Sidaard**, Michael Mongin, Osama A. AlSattam, and Keigo Hirakawa. "Comparison of Event Camera Processing Algorithms for Experimental 2D2C Velocimetry." In *AIAA SCITECH 2025 Forum*, p. 0475. 2025. <https://doi.org/10.2514/6.2025-0475>
66. Cai, Jielong, and **Sidaard Gunasekaran**. "Prediction of the Tandem Propeller Performance Using Linear Superposition Method." In *AIAA SCITECH 2025 Forum*, p. 1268. 2025. <https://doi.org/10.2514/6.2025-1268>
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68. Greenwood, Brock A., Abdul Khan, and **Sidaard Gunasekaran**. "Characterization of Agricultural Nozzle Spray Geometry in Propeller Wake." In *AIAA SCITECH 2025 Forum*, p. 2578. 2025. <https://doi.org/10.2514/6.2025-2578>
69. Pabon, Julian A., Grace Schreyer, **Sidaard Gunasekaran**, Michael Mongin, Aaron Altman, and Patrick Hammer. "Experimental Investigation of Prandtl-D3 Near Wake Signature." In *AIAA SCITECH 2025 Forum*, p. 2758. 2025. <https://doi.org/10.2514/6.2025-2758>

70. Porterfield, Andrew, **Sidaard Gunasekaran**, and Michael Mongin. "Closed-Loop Control of a Wing in Discrete Vortical Gust Encounter." In *AIAA SCITECH 2025 Forum*, p. 2762. 2025. <https://doi.org/10.2514/6.2025-2762>
71. Parlett, Samuel L., Jielong Cai, **Sidaard Gunasekaran**, and Michael Ol. "Experimental Investigation of a Variable Pitch Propeller at Various Incidence Angles in Foward Flight." In *AIAA SCITECH 2025 Forum*, p. 1267. 2025. <https://doi.org/10.2514/6.2025-1267>
72. Schreyer, Grace A., **Sidaard Gunasekaran**, Julian A. Pabon, and Jielong Cai. "Variations in the Wake Structure of Non-Elliptical Lift Distributions Near Wingtip." In *AIAA SCITECH 2025 Forum*, p. 0253. 2025. <https://doi.org/10.2514/6.2025-0253>
73. Kastner, Joseph R., **Sidaard Gunasekaran**, and Taber Wanstall. "Morphological Transitions in Droplet Breakup Under Shock-Induced Flows." In *AIAA SCITECH 2026 Forum*, p. 0006. 2026. <https://doi.org/10.2514/6.2026-0006>
74. Cain, Charles B., and **Sidaard Gunasekaran**. "Wake Characteristics of a Bell-Shaped Lift Distribution." In *AIAA SciTech 2026 Forum*, p. 2327. 2026. <https://doi.org/10.2514/6.2026-2327>
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76. Khan, Abdul R., and **Sidaard Gunasekaran**. "Sensitivity Analysis of Event Based Algorithms for Velocimetry." In *AIAA SCITECH 2026 Forum*, p. 1316. 2026. <https://doi.org/10.2514/6.2026-1316>
77. Schreyer, Grace A., Grace N. Selm, Julian A. Pabon, and **Sidaard Gunasekaran**. "Experimental Investigations of a Dual-Mode Skin-Actuated-Camber with Embedded Twist (SACET) Morphing Wing." In *AIAA SciTech 2026 Forum*, p. 0256. 2026. <https://doi.org/10.2514/6.2026-0256>
78. Parlett, Samuel L., Jielong Cai, Michael V. OL, and **Sidaard Gunasekaran**. "Experimental Investigation of a Variable Collective Pitch Propeller in Near Edgewise Flight." In *AIAA SCITECH 2026 Forum*, p. 0026. 2026. <https://doi.org/10.2514/6.2026-0026>
79. Pabon, Julian A., and **Sidaard Gunasekaran**. "Experimental Investigation of Dynamic Pitching Effects on a Delta Wing with Blown Jet." In *AIAA SCITECH 2026 Forum*, p. 0285. 2026. <https://doi.org/10.2514/6.2026-0285>
80. Cain, Charles B., Jessica C. DeMoor, and **Sidaard Gunasekaran**. "Cross-Stream PIV Characterization of the Prandtl-D3C Wake." In *AIAA SCITECH 2026 Forum*, p. 2328. 2026. <https://doi.org/10.2514/6.2026-2328>
81. Greenwood, Brock A., Abdul Khan, Megan Bender, and **Sidaard Gunasekaran**. "Characterization of Rotary Atomizer Spray Geometry Under the Influence of a Propeller." In *AIAA SCITECH 2026 Forum*, p. 2700. 2026. <https://doi.org/10.2514/6.2026-2700>

Non Peer-Reviewed Conference Articles

1. Porterfield, Andrew, Andrew Killian, and **Sidaard Gunasekaran**. "Generation and Characterization of Discrete Vortical Gust." In *2023 Regional Student Conferences*, p. 71873. 2023. <https://doi.org/10.2514/6.2023-71873>
2. Duncan, Lucas A., Jielong Cai, and **Sidaard Gunasekaran**. "Powered Wing Response to Periodic Gust Encounters." In *2023 Regional Student Conferences*, p. 71287. 2023. <https://doi.org/10.2514/6.2023-71287>

3. Jestus, Nevin, and **Sidaard Gunasekaran**. "Aerodynamic Interactions among Three Identical Wings in Close Proximity." In *2023 Regional Student Conferences*, p. 71488. 2023. <https://doi.org/10.2514/6.2023-71488>
4. Pabon, Julian A., Xinyu Gao, Jielong Cai, and **Sidaard Gunasekaran**. "Experimental Investigation of a Novel Morphing Wing Design." In *2023 Regional Student Conferences*, p. 71488. 2023. <https://doi.org/10.2514/6.2023-71704>

CITATIONS

As of Jan 2026, the overall citation count of scholarly work is 343. [Sidaard Gunasekaran - Google Scholar](#)

PATENTS

1. **Gunasekaran, Sidaard**, Julian Pabon, and Jielong Cai. "Skin actuated morphing wing." U.S. Patent 12,466,540, issued November 11, 2025.

RESEARCH PRESENTATIONS

At the Annual AIAA Conference:

All the peer-reviewed conference papers mentioned in research publications are accompanied by a 20-30 min presentations.

At the Dayton-Cincinnati Annual Science Symposium (DCASS)

1. Steven Goodman and Sidaard Gunasekaran, "*Investigation of Near Wake Turbulent Fluctuations and its Relation to Wing Performance*", [DCASS 2018](#), Sinclair Community College, Dayton, OH.
2. Daniel Curry and Sidaard Gunasekaran, "*Effect of Segmented Trailing Edge Extensions on Aerodynamic Efficiency*", [DCASS 2018](#), Sinclair Community College, Dayton, OH.
3. Nathan Thomas and Sidaard Gunasekaran, "*Affecting the Wingtip Vortex by Influencing Wing Surface-Flow Direction*", [DCASS 2018](#), Sinclair Community College, Dayton, OH.
4. Thomas Cook and Sidaard Gunasekaran, "*Characterization of Aerotech Shuttering System*", [DCASS 2019](#), Sinclair Community College, Dayton, OH.
5. Jielong Cai and Sidaard Gunasekaran, "*Changes in Propeller Performance Due to Ground Proximity*", [DCASS 2019](#), Sinclair Community College, Dayton, OH.
6. Faith Loughnane, Rachel Supina, and Sidaard Gunasekaran, "*Wing Performance Changes Due to Wing Surface Contours*", [DCASS 2019](#), Sinclair Community College, Dayton, OH.
7. Josh Deslich and Sidaard Gunasekaran, "*Effect of Slotted Winglet on the Wingtip Vortex*", [DCASS 2019](#), Sinclair Community College, Dayton, OH.
8. Neal Novotny, Jielong Cai, and Sidaard Gunasekaran, "*Experimental Investigation of a Spanwise Axis Turbine in a Symmetric Wing*", [DCASS 2019](#), Sinclair Community College, Dayton, OH.
9. Samuel Barnhart and Sidaard Gunasekaran, "*Investigation of Scaled Down Doppler LIDAR for Velocity Measurements in Wind Tunnels*", [DCASS 2019](#), Sinclair Community College, Dayton, OH.

10. Benjamin Moidel, Tim Reissman, Megan Reissman, Sidaard Gunasekaran, *“Modeling and Validation of Pilot-Aircraft Cybernetic Systems”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
11. Samuel Barnhart and Sidaard Gunasekaran, *“Investigation of Doppler LIDAR for Velocity Measurements in Wind Tunnels”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
12. Rachael Supina and Sidaard Gunasekaran, *“Tailoring Lift Distribution Through Trailing Edge Extension Twist”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
13. Lucas Hung, Jielong Cai, Neal Novotny, and Sidaard Gunasekaran, *“Development of a Multi-Purpose Flap and Spoiler Mechanism for High Endurance Unmanned Aerial Vehicles”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
14. Faith Loughnane, Michael Mongin, and Sidaard Gunasekaran, *“Effect of Airfoil-Preserved Undulations on Wing Performance”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
15. Samuel Barnhart and Sidaard Gunasekaran, *“Lift Coefficient Prediction of Distributed Blown Wings”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
16. Michael Mongin and Sidaard Gunasekaran, *“Frequency Response of a Shuttered Open Jet Wind Tunnel”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
17. Rachel Sharp and Sidaard Gunasekaran, *“Turbulent Near Wake Properties at Maximum Aerodynamic Efficiency Condition”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
18. Jielong Cai, Michael Ol, Anwar Ahmed, and Sidaard Gunasekaran, *“Propeller Partial Ground Effect”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
19. Scott Chris, Matt Gazella, Jonathan Hill, Austin Abel, and Sidaard Gunasekaran, *“Design and Analysis of an Additive Manufactured Supersonic Wind Tunnel”*, [DCASS 2020](#), Sinclair Community College, Dayton, OH.
20. Jielong Cai, Michael Ol, and Sidaard Gunasekaran, *“Changes in RC Propeller Performance under Various Mounting Pitch Angles”*, [DCASS 2021](#), Sinclair Community College, Dayton, OH.
21. Grace Culpepper, Jielong Cai, and Sidaard Gunasekaran, *“Propeller and Propeller-In-Wing Thrust Vectoring”*, [DCASS 2021](#), Sinclair Community College, Dayton, OH.
22. Andrew Killian, Michael Mongin, Albert Medina, and Sidaard Gunasekaran, *“Design and Characterization of Vortical Gust Generator in the University of Dayton Water Tunnel”*, [DCASS 2022](#), Sinclair Community College, Dayton, OH.
23. Nevin Jestus and Sidaard Gunasekaran, *“Wing-to-Wing Interactions on Distributed Lift Applications”*, [DCASS 2022](#), Sinclair Community College, Dayton, OH.
24. Jielong Cai and Sidaard Gunasekaran, *“Propeller Ground and Ceiling Effect in Forward Flight”*, [DCASS 2022](#), Sinclair Community College, Dayton, OH.
25. Ian Tierney and Sidaard Gunasekaran, *“Wake Effects on Agricultural Spray Using Drones”*, [DCASS 2022](#), Sinclair Community College, Dayton, OH.

26. Nevin Jestus and Sidaard Gunasekaran, "*Aerodynamic Interactions between Three Identical Wings in Close Proximity*", [DCASS 2023](#), Sinclair Community College, Dayton, OH.
27. Michael Mongin, Aaron Altman, Nevin Jestus, Sidaard Gunasekaran, "*Extended High Lift Characteristics of Distributed Lift Configurations*", [DCASS 2023](#), Sinclair Community College, Dayton, OH.
28. Jielong Cai and Sidaard Gunasekaran, "*Vertically Offset Overlapping Propellers in Tandem*", [DCASS 2023](#), Sinclair Community College, Dayton, OH.
29. Jielong Cai and Sidaard Gunasekaran, "*Sinusoidal Gust Response of RC Propellers in Tandem Configuration*", [DCASS 2023](#), Sinclair Community College, Dayton, OH.
30. Luke Duncan and Sidaard Gunasekaran, "*Propeller Wing Interaction and Response to a Sinusoidal Gust*", [DCASS 2023](#), Sinclair Community College, Dayton, OH.
31. Andrew Killian, Michael Mongin, Albert Medina and Sidaard Gunasekaran, "*A Comparison of Unsteady Aerodynamic Models to Experiments in Vortical Gust Encounters and Closed Loop Mitigation*", [DCASS 2023](#), Sinclair Community College, Dayton, OH.
32. Pabon, Julian A., Xinyu Gao, Jielong Cai, and Sidaard Gunasekaran. "Experimental Investigation of a Novel Morphing Wing Design.", [DCASS 2024](#), Sinclair Community College, Dayton, OH.
33. Jielong Cai and Sidaard Gunasekaran, "*Positively Staggered Vertically Offset Propellers in Tandem*", [DCASS 2024](#), Sinclair Community College, Dayton, OH.
34. Abdul Khan, Brock Greenwood, and Sidaard Gunasekaran, "Characterization of Nozzle Spray Geometry in Propeller Wake", [DCASS 2024](#), Sinclair Community College, Dayton, OH.
35. Michael Foster, Jessica DeMoor, Jielong Cai, and Sidaard Gunasekaran, "Flight Test Validation of Tandem Propeller Performance with Vertical and Horizontal Offset", [DCASS 2024](#), Sinclair Community College, Dayton, OH.
36. Porterfield, Andrew, Andrew Killian, Sidaard Gunasekaran, and Michael Mongin. "Discrete Vortical Gust Encounter and Mitigation using Closed Loop Control." [DCASS 2024](#), Sinclair Community College, Dayton, OH.
37. Emma Schutter, Jielong Cai, and Sidaard Gunasekaran, "On the Linear Superposition of Wing and Propeller Performance in a Wing Embedded Propeller System", [DCASS 2024](#), Sinclair Community College, Dayton, OH.
38. Foster, Michael, Jessica DeMoor, Sidaard Gunasekaran, and Jielong Cai. "Flight Test Validation of Tandem Propeller Performance With Vertical Offset." [DCASS 2025](#), Sinclair Community College, Dayton, OH.
39. Greenwood, Brock A., Abdul Khan, and Sidaard Gunasekaran. "Characterization of Agricultural Nozzle Spray Geometry in Propeller Wake." [DCASS 2025](#), Sinclair Community College, Dayton, OH.
40. Pabon, Julian A., Grace Schreyer, Sidaard Gunasekaran, Michael Mongin, Aaron Altman, and Patrick Hammer. "Experimental Investigation of Prandtl-D3 Near Wake Signature." [DCASS 2025](#), Sinclair Community College, Dayton, OH.

41. Schreyer, Grace A., Sidaard Gunasekaran, Julian A. Pabon, and Jielong Cai. "Variations in the Wake Structure of Non-Elliptical Lift Distributions Near Wingtip." [DCASS 2025](#), Sinclair Community College, Dayton, OH.

42. Cain, Charles B., and Sidaard Gunasekaran. "Wake Characteristics of a Bell-Shaped Lift Distribution." [DCASS 2025](#), Sinclair Community College, Dayton, OH.

At the Dayton Engineering Student Science Symposium (DESS)

32. Steven Goodman and Sidaard Gunasekaran, "*Investigations of Near Wake Turbulent Fluctuations and its Relation to Wing Performance*", [DESS 2017](#), Wright State University, Dayton, OH.

33. Neal Novotny and Sidaard Gunasekaran, "Design and Characterization of Wind Lens Grid Array", [DESS 2018](#), Wright State University, Dayton, OH.

34. Rachael Supina, Michael Mongin, and Sidaard Gunasekaran, "*Wing Performance Changes Due to Trailing Edge Extensions*", [DESS 2019](#), Wright State University, Dayton, OH.

35. Faith Loughnane, Rachael Supina, Michael Mongin, and Sidaard Gunasekaran, "*Effect of Airfoil-Preserved Undulations on Wing Performance*", [DESS 2019](#), Wright State University, Dayton, OH.

36. Jielong Cai, and Sidaard Gunasekaran, "*Propeller Partial Ground Effect*", [DESS 2019](#), Wright State University, Dayton, OH.

37. Scott Chris, Matt Gazella, Jonathan Hill, Austin Abel, and Sidaard Gunasekaran, "*Design and Analysis of an Additive Manufactured Supersonic Wind Tunnel*", [DESS 2019](#), Wright State University, Dayton, OH.

38. Samuel Barnhart and Sidaard Gunasekaran, "*Investigation of Scaled Down Doppler LIDAR for Velocity Measurements in Wind Tunnels*", [DESS 2019](#), Wright State University, Dayton, OH.

39. Neal Novotny and Sidaard Gunasekaran, "*Performance and Proximity Investigations on Small Scale Lensed Turbines*", [DESS 2019](#), Wright State University, Dayton, OH.

40. C.B.Cain and Sidaard Gunasekaran, "*Wake Characteristics of Non-Elliptical Lift Distribution*", [DESS 2024](#), University of Dayton, Dayton, OH.

41. Sidaard Gunasekaran, "*Evaluation of Event Cameras for 2D2C Velocimetry*", [DESS 2024](#), University of Dayton, Dayton, OH.

RESEARCH SUPPORT

Acquired External Funding: (Total: \$740,747)

1. Equipment Grant for Custom Agricultural Spray Drift Test Section for University of Dayton Low Speed Wind Tunnel, **Spray Analytics Incorporated**, 08/15/2018, \$25,000.

2. Enhanced Optical Flow Diagnostics/Minimization of Particle Seeding, **Dayton Area Graduate Student Institute (DAGSI/AFRL)**, 08/15/2018 – 08/14/2019, \$63,831.

3. Massively Unsteady, Vortex Dominated Flows on Pitching and Plunging Wings, **Dayton Area Graduate Student Institute (DAGSI/AFRL)**, 08/15/2021 – 08/14/2022, \$58,810.

4. Application of Traditional and Enhanced Machine Learning and Data Science for Active Flow Control Techniques, **Defense Advanced Research Projects Agency (DARPA), Control of Revolutionary Aircraft with Novel Effectors (CRANE)**, UD Seedling Effort, 08/15/2021 – 05/15/2022, \$70,000.
5. Effect of Propeller Wake on Drift Potential of Agricultural Sprays, **Badische Anilin and Soda Fabrik (BASF)**, 01/15/2022 – 08/15/2022, \$50,000.
6. Wind Tunnel Force Based Testing on CRG Wing Models, 2/13/2023 – 2/25/2023, **Corner Stone Research Group (CRG)**, \$24,703 (**J943C0**)
7. **2023 KEEN National Rising Star Award**, \$25,000 (**414004 – 220211 – A – 200000 – 100202 - 999999**)
8. Evaluation of Event Cameras for Velocimetry, **Air Force Office of Scientific Research (AFOSR)**, 09/11/2023 – 09/10/2024, \$66,978 (**MXA011**)
9. Evaluation of Event Cameras for 2D3C Velocimetry, **Air Force Office of Scientific Research (AFOSR)**, 01/15/2025 – 12/15/2026, \$77,765 (**N4W000**)
10. DURIP: Benchmarking Stereo Event-Based Velocimetry for 3D Flowfields Using Lagrangian Reference Measurements, **Air Force Office of Scientific Research (AFOSR)**, - \$284,136, GRANT14409574, (**Funding Approved**)
11. Funding Acquired through Wind Tunnel Testing for Industries

A. Spray Analytics

1. Dec 19 th , 2018	-	\$812.50
2. Feb 12 th , 2019	-	\$812.50
3. Aug 25 th , 2020	-	\$1,200.00
4. Nov 6 th , 2020	-	\$1,200.00
5. Nov 16 th , 2021	-	\$3,000.00
6. Oct 1 st – 4 th , 2024	-	\$7,500.00
7. Dec 8 th – 10 th , 2025	-	\$5,000.00

B. Inteva Products

1. July 9 th , 2021	-	\$1029.33
2. Oct 25 th , 2021	-	\$789.153

C. Cornerstone Research Group

1. Feb, 2023	-	\$24,000
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Acquired Internal Funding: (Total: \$364,700)

1. Revision of Introduction to Flight Course, **University of Dayton Hanley Sustainability Institute (HSI)**, 05/16/2017 – 08/15/2017, \$2,500.
2. Affecting Drag of a Wing through Periodic Actuation of Smart Material in its Wake, **Research Council Seed Grant (RCSG)**, 05/15/2017 – 08/15/2017, \$5,000.

3. Creativity Across the Curriculum Fellowship Cohort Program, **University of Dayton Institute of Applied Creativity for Transformation**, 08/15/2017 – 05/14/2018, \$5,000.
4. Use of Trailing Edge Extensions for Enhanced Propeller Performance. University of Dayton **Clare Boothe Luce Research Scholars Program**, PI, Faith Loughnane (BS MEE student), 08/15/2018 – 05/15/2020, \$25,200.
5. Product Design and Development of Lensed Turbines, **Leonardo Enterprises**, 05/16/2019 – 08/15/2019, \$10,000.
6. Thrust vectoring characteristics of Wing Embedded Propeller. University of Dayton **Clare Boothe Luce Research Scholars Program**, PI, Grace Culpepper (BS MEE student), 08/15/2019 – 05/15/2021, \$25,200.
7. Deep Learning Algorithm for Atomization Characterization using Shadowgraph Images: A Novel Approach for Spray Drift Analysis, **University of Dayton STEM Catalyst**, Sidaard Gunasekaran (UD - PI), Barath Narayanan (UDRI - Co-PI), 05/16/2020 – 08/16/2021, \$50,000.
8. Event-based PIV for flow visualization in laser ablation processes, UDRI Fellowship, 05/16/2025 – 08/16/2025, \$15,000. (1.5 months of summer salary)

Funding Acquired from Kern Family Foundation: (Total: \$141,889)

1. Incorporating EML in Fundamental Aerodynamic Course. **KEEN Cohort III**, Part of UD's institutional KEEN grant, 05/16/2017 – 08/15/2017, \$13,500.
2. Review of EML Cards in EngineeringUnleashed.com. **Community Catalyst Program, Kern Entrepreneurial Engineering Network**, 01/15/2018 – 05/15/2018, \$2,500
3. Review of EML Cards in EngineeringUnleashed.com. **Community Catalyst Program, Kern Entrepreneurial Engineering Network**, 08/15/2018 – 12/05/2018, \$2,500
4. Design and Implementation of EML Modules in Aerospace Curriculum. **Innovative Curriculum with Entrepreneurial Mindset (ICE) Workshop, Kern Family Foundation**, 01/15/2018 – 08/15/2018, \$2,000.
5. Review of EML Cards in EngineeringUnleashed.com. **Community Catalyst Program, Kern Entrepreneurial Engineering Network**, 01/15/2019 – 05/15/2019, \$2,500
6. Review of EML Cards in EngineeringUnleashed.com. **Community Catalyst Program, Kern Entrepreneurial Engineering Network**, 08/15/2019 – 12/05/2019, \$2,500
7. KEEN Summer Immersion Program, “Developing Course Modules for Aerodynamics with Industry Partnership”, **KERN Family Foundation** 5/15/2019 – 8/15/2019, \$5,000.
8. Review of EML Cards in EngineeringUnleashed.com. **Community Catalyst Program, Kern Entrepreneurial Engineering Network**, 01/15/2020 – 05/15/2020, \$2,500
9. KEEN Summer Immersion Program, “Developing Course Modules for Aerodynamics with Industry Partnership”, **Kern Entrepreneurial Engineering Network** 5/15/2020 – 8/15/2020 \$5,000.
10. Creating EM-infused CourseDecks for Aerodynamics and Experimental Aerodynamics on Engineering Unleashed, **Kern Entrepreneurial Engineering Network**, 05/15/2021 – 08/15/2021, \$10,000.

11. Creating EM-infused CourseDecks for Aerodynamics and Experimental Aerodynamics on Engineering Unleashed, **Kern Entrepreneurial Engineering Network**, 05/15/2021 – 08/15/2021, \$10,000.
12. Designing New Partner On-boarding Toolkit for Ambassadors, Kern Entrepreneurial Engineering Network, 08/15/2022 – 05/15/2023, \$27,720. **(KEEN)**
13. Building iFly Case Study for Aerodynamics, Fluid Dynamics, and Statics, Kern Entrepreneurial Engineering Network, 05/15/2023 – 08/15/2023, \$5,000. **(KEEN)**
14. Serving as an Departmental Community Catalyst, Kern Entrepreneurial Engineering Network, 05/15/2023 – 08/15/2023, \$2,500. **(KEEN)**
15. Developing iFLY Case Studies, 08/15/2023 – 05/15/2024, \$19,335. **(KEEN)**
16. Design and Dissemination of Case Studies and Process for Developing Case Studies, 08/15/2024 – 12/15/2024, \$9,667. **(KEEN)**
17. EMxAI summit and roadshow design and planning, 01/15/2026 – 05/15/2026, \$9,667. **(KEEN)**
18. 2025 Engineering Unleashed Fellow, \$10,000

Workshop Facilitation: Total (\$13,500)

1. Developing Crescendo Workshops, Kern Family Foundation, Waukesha, WI, April 7th – 9th, 2025, \$2,500
2. KEEN EM Observable Behaviors Pilot Workshop, Milwaukee School of Engineering (MSOE), June 1st – June 3rd, 2025. \$1,500
3. Crescendo Workshop: Mastering Modern Academia, A Workshop on AI Strategies for Faculty Success with EM, University of Toledo, June 15th, 2025, \$2,500
4. Crescendo Workshop: Mastering Modern Academia, A Workshop on AI Strategies for Faculty Success with EM, Trine University, Aug 13th, 2025, \$2,500
5. Crescendo Workshop: The Intentional Joy of Learning, A Workshop on Curiosity, Agency, and Student Engagement, New York Institute of Technology, Nov 5th, 2025, \$2,500
6. Increasing Faculty Engagement with EM using KEEN Crescendo Workshops, University of Cincinnati, Dec 16th, 2025. \$2,000

Course Release

1. “What is the University For?”, 2018 Faculty Seminar, **University of Dayton**. Course Buyout for Fall 2018. (Equivalent Cost: \$10,250 (without benefits))

Travel Funds – Total (\$1,500)

1. Funding Support for Students to Travel to SCITECH 2021 conference in San Diego, **Local Chapter (Dayton/Cincinnati) of American Institute of Aeronautics and Astronautics (AIAA)**, 12/05/2021, \$500.

2. Funding Support for Students to Travel to SCITECH 2022 conference in Washington DC, **Local Chapter (Dayton/Cincinnati) of American Institute of Aeronautics and Astronautics (AIAA)**, 12/05/2022, \$500.
3. Funding Support for Students to Travel to SCITECH 2023 conference in Orlando, FL, **Local Chapter (Dayton/Cincinnati) of American Institute of Aeronautics and Astronautics (AIAA)**, 12/05/2023, \$500.

Funds for Student Clubs

1. K30 Jet Engine from KingTech for the Design, Build, and Fly (DBF) Student Club, \$1,590, Spring 2019
2. Ultem 1010 3D Print Material from AFRL RQH High Speed Systems Division, for American Institute of Aeronautics and Astronautics (AIAA) student club, \$5,000, Spring 2020
3. Sponsor funding for hosting 2023 Region III Student Conference at the University of Dayton, \$6,000, Spring 2023

FEATURED RESEARCH WORK:

1. Gail C. Keck | Jun 07. (2019, June 7). *Spraying in the Ohio wind*. Farm Progress. Retrieved June 18, 2022, from <https://www.farmprogress.com/crops/spraying-ohio-wind>
2. University of Dayton Wind Tunnel Used For Agricultural Chemical, Equipment Testing, **ASEE First Bell**, Jun 10, 2019
3. University of Dayton. (2022, March 11). *Using Machine Learning to Minimize Spray Drift*. YouTube. Retrieved June 18, 2022, from <https://www.youtube.com/watch?v=zZs8T2pB2Os>
4. *Low-speed wind tunnel*. Projects : University of Dayton, Ohio. Retrieved June 18, 2022, from <https://udayton.edu/engineering/research/research-labs/wind-tunnel/projects.php>
5. Keck, Gail. "Researchers Study Drone Spray Drift Potential." Farm and Dairy, 12 Apr. 2023, Accessed January 25, 2024 www.farmanddairy.com/news/researchers-study-drone-spray-drift-potential/766280.html.
6. Griffiths, Clinton. 2023. "Rise of the Spray Drone." AgWeb. August 31, 2023. Accessed January 25, 2024 <https://www.agweb.com/news/crops/crop-production/rise-spray-drone>.
7. "News." University of Dayton research-industry partnership seeks answers to precision agriculture : University of Dayton, Ohio, January 19, 2024. Accessed January 25, 2024 https://udayton.edu/news/articles/2023/04/sid_gunasekaran_spray_analytics_drone_testing.php
8. Thompson, Ann. "Agriculture Drones Are Getting More Popular. Are They Accurate Enough to Spray Crops?" WVXU. Accessed January 25, 2024. <https://www.wvxu.org/podcast/focus-on-technology/2023-04-10/agriculture-drones-crops-spray-drift>.
9. Schwartz, David. n.d. "U Dayton Industry Partnership Looks to Advance Drone Spraying in Precision Agriculture - University-Industry Engagement Week - Tech Transfer Central." Tech Transfer Central.



Accessed January 25, 2024 <https://techtransfercentral.com/2023/04/11/u-dayton-industry-partnership-looks-to-advance-drone-spraying-in-precision-agriculture/>.

INVITED TALKS:

1. **Gunasekaran, Sidaard.** *Aerospace Research – Aero Problems*, University of Dayton Junior/Senior seminar. 2015.
2. **Gunasekaran, Sidaard.** *“Errors in Off-axis Loading of Off-the-shelf 6-Component Force Transducers: A Cautionary Tale”*, Michigan State University, Michigan. October 2017.
3. **Gunasekaran, Sidaard.** *“Frequency Response of Shuttered Open Jet Wind Tunnel”*, Subsonic Aerodynamic Testing Association (SATA) Conference, Dallas, TX, May 2022.
4. **Gunasekaran, Sidaard.** *“Measuring Droplet Size Distribution in the Propeller Wake: A Cautionary Tale”*, Spray Drift Seminar, University of Dayton, OH, May 2023
5. **Gunasekaran, Sidaard.** *“Characterizing Flow Angularity in a Shuttered Open Jet Wind Tunnel using Phase-Locked PIV”*, Subsonic Aerodynamic Testing Association (SATA) Conference, United States Air Force Academy (USAF), Colorado Springs, June 2023.
6. **Gunasekaran, Sidaard.** *Increasing Faculty Engagement with EM using KEEN Crescendo Workshops*, University of Cincinnati, Dec 16th, 2025.

WORKSHOPS AND KEYNOTES

1. *EM Essentials, Introduction to Entrepreneurial Mindset*, 2024 KEEN National Conference, Austin, TX. <https://kernfamilyfdn.eventsair.com/2024-keen-national-conference/agenda>
2. *Development of Case Studies in Partnership with iFLY Skydiving Wind Tunnel*, Keynote Speaker and Panelist, 2024 KEEN National Conference, Austin, TX. <https://www.youtube.com/watch?v=xQYK3G58xOk&t=1193s>
3. *Creating Value and Connections with Writing-To-Learn in STEM*, 2024 KEEN National Conference, Austin, TX. <https://kernfamilyfdn.eventsair.com/2024-keen-national-conference/agenda>
4. *Transform your Teaching with Case Studies*, 2024 ASEE Annual Conference, Portland, OR. https://nemo.asee.org/public/conferences/344/registration/view_session?session_id=19447
5. *Transform your Teaching with Case Studies*, 2025 KEEN National Conference, Austin, TX. <https://kernfamilyfdn.eventsair.com/2025-keen-national-conference/agenda>
6. *Crescendo Workshop: Mastering Modern Academia*, A Workshop on AI Strategies for Faculty Success with EM, University of Toledo, June 15th, 2025
7. *Crescendo Workshop: Mastering Modern Academia*, A Workshop on AI Strategies for Faculty Success with EM, Trine University, Aug 13th, 2025
8. *Mastering Modern Academia*, A Workshop on AI Strategies for Faculty Success with EM, KEEN Leader Meeting, Milwaukee, Sept 25th, 2025
9. *Crescendo Workshop: The Intentional Joy of Learning*, A Workshop on Curiosity, Agency, and Student Engagement, New York Institute of Technology, Nov 5th, 2025

10. Crescendo Workshop (Online): The Intentional Joy of Learning, A Workshop on Curiosity, Agency, and Student Engagement, University of Idaho, Dec 4th, 2025

POSTER PRESENTATIONS

1. **Gunasekaran, Sidaard**, and Aaron Altman. "Is There a Relationship Between the Wingtip Vortex and the Shear Layer?." Sigma-Xi Celebration of Science, University of Dayton, Dayton, OH. 2016.
2. **Gunasekaran, Sidaard**, and Aaron Altman. "Relationship between Wingtip Vortex, Free Shear Layer and Aerodynamic Efficiency", Graduate Student Showcase, University of Dayton, Dayton, OH. 2016.
3. Thomas, Nathan and **Sidaard Gunasekaran**. "Affecting Aerodynamic Efficiency by Influencing Wing Surface-Flow Direction.", Summer Undergraduate Research Experience, University of Dayton, Dayton, OH. 2017.
4. Palmer, Asa and **Sidaard Gunasekaran**. "Effect of Curved Boundary Layer Fences on Aerodynamic Efficiency", Ohio Space Grant Consortium, Cleveland, Ohio. 2018.
5. **Gunasekaran, Sidaard**. "Making a Tangible Different Through Fundamental Aerodynamic Research", GEM-ASEE Engineering Research Showcase, Washington, D.C. 2018.

LABORATORY DEVELOPMENT

University of Dayton Low Speed Wind Tunnel (UD-LSWT)

Partnered with Spray Analytics (<https://sprayanalytics.com/>) to get the University of Dayton Low Speed Wind Tunnel (UD-LSWT) certified by the Environmental Protection Agency (EPA) for conducting spray drift measurements and drift reduction technology testing. The UD-LSWT is now the second non-government wind tunnel to perform such experiments. Developed a cost model for the UD-LSWT which allows companies such as Monsanto, DelAgro, Warrant, Powermax, Enlist, Reign, Intact, Interlock, etc. perform testing at the UD-LSWT for a charge of \$325 per hour.

University of Dayton Water Tunnel (UDWaT)

University of Dayton Water Tunnel (UD-WaT) was established as one of the aerospace labs in the mechanical and aerospace engineering department in Spring of 2020. The water tunnel offers a range of Reynolds number to test which is far below what the UD-LSWT can offer. Unlike the UD-LSWT, the water tunnel is predominantly a research facility established in collaboration with AFRL, and fitted with equipment to design, develop, and test closed loop control system architectures to mitigate unsteady and gust loads on wings. The closed loop control system developed in the lab allows for lift tracking and control over a range of reduced frequencies and amplitudes, especially when subjected to external discrete and periodic disturbances. It also allows testing of many of the existing unsteady models for its robustness and accuracy.

STUDENT CLUB MENTORING

1. Design, Build, and Fly (DBF) Student Club – Faculty Advisor (2017 – Present)

Won 1st place in the OSGC sponsored solar power competition, 2018

- *May 2018 UAS Competition - OSGC UAS*. Google Sites: Sign-in. Retrieved June 18, 2022, from <https://sites.google.com/site/osgcstars/home/competitions>

2. IT Flies Student Club – Faculty Advisor (2017 – Present)

Organized the following competitions –

2018 IT Flies competition

- YouTube. (2018, May 18). *It flies competition 2018 Slideshow - University of Dayton*. YouTube. Retrieved June 18, 2022, from <https://www.youtube.com/watch?v=6D7UezMD92k&t=87s>

2019 IT Flies competition

- YouTube. (2019, April 18). *It flies competition 2019 Slideshow - University of Dayton*. YouTube. Retrieved June 18, 2022, from <https://www.youtube.com/watch?v=s4qkH90wGBk&t=4s>

2021 IT Flies competition

- *IT Flies! Merlin Flight Competition*. Merlin Flight Simulator Competition : University of Dayton, Ohio. (n.d.). Retrieved June 18, 2022, from https://udayton.edu/engineering/departments/mechanical_and_aerospace/it-flies.php

2022 IT Flies competition

- Updyke, K. *University of Dayton Team Takes First Place at It Flies International Flight Simulation Competition*. University of Dayton team takes first place at IT Flies international flight simulation competition : University of Dayton, Ohio. Retrieved June 18, 2022, from <https://udayton.edu/blogs/engineering/2022/22-05-23-itflies.php>

2023 IT Flies competition

- Merlin Flight Simulator Competition : University of Dayton, Ohio.” n.d. Udayton.edu. Accessed January 25, 2024. https://udayton.edu/engineering/departments/mechanical_and_aerospace/it-flies.php.
- “IT FLIES USA - Flight Simulator, Flight Simulators for Academic, Engineering and Pilot Training - Merlin Flight Simulation Group.” n.d. Www.merlinsim.com. Accessed January 25, 2024. <http://www.merlinsim.com/competitionusa.htm>.
- “IT FLIES USA 2023”, Aerospace Magazine, September 2023 Edition, Page 51, [aerospace-magazine-september-2023.pdf \(aerosociety.com\)](#)

2024 IT Flies Competition

- [Test-Pilots-Fly-and-Assess--Student-Teams-Aircraft-Designs-at-IT-FLIES-USA \(ukexportnews.co.uk\)](#)
- [Merlin Flight Simulator Competition : University of Dayton, Ohio \(udayton.edu\)](#)

3. American Institute of Aeronautics and Astronautics (AIAA) Student Club – Faculty Advisor (2016 – Present)

Organized the following events –

- Fall 2016 – Invited [Mr. Eugene Fleeman to UD](#) to speak about Missile Design, Development and System Engineering
- Fall 2017 – Invited [Mr. Tom Morgenfeld, the X-35 Test pilot to UD](#) to speak about Development of the X-35 prototype and Test Flight. Below is a picture during the event.

- Spring 2018 – Assisted local chapter with dinner meeting with [Mr. Kenneth Szalai as a guest speaker](#) to speak about Flights of Discovery – Impact and Lessons from Experimental Flight Research at NASA Neil Armstrong Flight Research Center at the Four Bridges Country Club.
- Spring 2019 – Invited [Mrs. Michelle Evans, the author of X-15 Rocket plane to UD](#) to talk about the X-15 rocket plane, flying the first wings to space. Below are some pictures from the event
- Fall 2019 – Invited [Mr. Pradeep Raj to UD](#) to talk about Applied Computational Aerodynamics (ACA), Reflections on Long and Exciting Journey in the never-ending pursuit of effectiveness
- Spring 2022 – Invited [Mr. Walter Gordon to UD](#) to talk about Project Gemini: Unsung Hero of Neil's Armstrong's One Small Step
- Spring 2023 – Organized and Hosted [AIAA Region III Student Conference](#)

PROFESSIONAL SERVICE AND OUTREACH ACTIVITIES

- 2025 – **Member of 2025 KEEN Rising Star Committee**
- 2025 - **Member of 2025 KEEN National Conference (KNC) Planning Committee and Lead Workshop Coach**
- 2024 – **Member of 2024 KEEN National Conference (KNC) Planning Committee and Lead Workshop Coach**
- 2023 – **Member of 2023 KEEN National Conference (KNC) Planning Committee and Lead Workshop Coach**
- 2023 – **Hosted Spray Drift Seminar at the University of Dayton**
- 2023 – **Hosted American Institute of Aeronautics and Astronautics (AIAA) Region III Student Conference at the University of Dayton and served in the planning committee**
- 2022 – **Member of 2023 KEEN National Committee (KNC) and Workshop Coaching**
- 2020 – **KEEN Subnet Catalyst**
 1. Helped revamp the use of subnets in the EU.com. The subnets were then changed to groups in the new version of the EU.com
- 2019 – 2020: **KEEN Ambassador**
 1. Helped kickstart the onboarding initiatives to join the KEEN network in the following institutions:
 - Valparaiso University
 - George Washington University
 - Drexel University
 - Florida Gulf Coast University
 - Duke University
 - Michigan Tech University
 - Rochester Institute of Technology
 - Clarkson University
 - Tulane University
 - University of Toledo
 2. Following activities were conducted in each institution:
 - **Strategic Doing Workshop:** A 2-hour workshop with 6-8 KEEN leaders driving the KEEN initiatives in respective institutions. The idea is to identify high impact and relatively easy to implement ideas to get KEEN initiatives moving forward.
 - **Overview of Kern Family Foundation and KEEN Network:** This 1.5-hour session was open to all faculty and covers KEEN, entrepreneurially minded learning (EML) and modules that I have developed in my courses, demonstrating the EngineeringUnleashed.com platform and candid discussions about the entrepreneurially minded learning movement.

- **Funding Opportunities:** A 30-minute discussion with faculty to talk about funding opportunities
- 2018 – 2020: **KEEN Community Catalyst**
 1. Helped the development of Engineering Unleashed.com (EU.com) since its inception. I have trained several faculties from different institution in creation and adoption of cards from EU.com.
- 2018 – Present: **Hosted Air Camp at the University of Dayton in Summer each year from 2018**
<https://bit.ly/AirCampUD2018>
- 2017 – Present: **Journal and Abstract Reviewer for**
 1. AIAA Journal (6)
 2. Journal of Aircraft (2)
 3. Flow Turbulence and Combustion (2)
 4. Vertical Flight Society (2)
 5. Journal of Aerospace Engineering (1)
 6. Aerospace Science and Technology (5)
 7. MDPI Journals (5)
 8. International Journal of Micro Air Vehicles (1)
 9. International Journal of Mechanical Engineering Education (3)
 10. American Society of Engineering Education (12)
 11. American Institute of Aeronautics and Astronautics SCITECH (8)
 12. National Science Foundation (NSF) (1)
- 2017 – Present: **Helped Organize Local AIAA events around Dayton** <https://bit.ly/AIAAService>
- 2017 – 2018: **Hosted Spray Drift Seminar at the University of Dayton** <https://bit.ly/SprayDriftSeminar>

OUTREACH AND SCIENCE COMMUNICATION

1. Featured on CET/ThinkTV's "I Can Be Anything I Want to Be, A to Z" introducing aerospace engineering to children. <https://atoz.thinktv.org/a.html>
2. Organized AIAA outreach at Dayton Air Show, Aero Carnival, TechFest, WACO Fly-In, teaching students through hands-on STEM demos. <https://bit.ly/AIAAService>
3. Advocate with Citizens for Space Exploration (2017–2019), meeting U.S. Senators and Members of Congress on NASA and STEM policy.

HONORS AND AWARDS

- 2025: KEEN EngineeringUnleashed Fellow
- 2025: UD-UDRI Summer Fellowship
- 2024: Hans von Ohain Endowed Chair
- 2024: DCASS – 1st Place in Art in Science Competition
- 2023: 2023 KEEN National Rising Star Award [Story: 2023 KEEN Rising Star: Sid G | Engineering Unleashed](#)
- 2023: 2023 KEEN Campus Rising Star Award
- 2023: AIAA Special Service Citation for Outstanding Service to AIAA and the Student Branch for the Region III Student Conference
- 2018: University of Dayton School of Engineering VISION award for Innovation in Teaching
- 2017: ASME Dayton Engineering Science Symposium Best Presentation in Fluid Dynamics Award
- 2016: University of Dayton Outstanding PhD Research Award

- 2015: American Institute of Aeronautics and Astronautics Chairman's Award for Outstanding Support of the AIAA Dayton-Cincinnati Section K-12 STEM Outreach Program.
- 2015: Graduate Summer Fellowship awarded by the University of Dayton
- 2014: Dayton-Cincinnati Aerospace Sciences Symposium Best Presentation Award (with Aaron Altman)
- 2013: Graduate Summer Fellowship awarded by the University of Dayton
- 2007: Academic Excellence Award from Sainik School Amaravathi Nagar (India)
- 2007: Best in Physics Award from Sainik School Amaravathi Nagar (India)

With Students:

- 2025: Charles Cain – ASME DESS Best Presentation Award
- 2023: Andrew Killian – DCASS 1st Place in Art-In-Science Competition <https://www.aiaa-daycin.org/DCASS/48dcass.php>
- 2023: Julian Pabon and Xinyu Gao (Cindy) – 2nd Place in Best Presentation in Undergraduate Category in AIAA Region III Student Conference, Dayton, OH. [News Detail | AIAA](#)
- 2023: Andrew Porterfield and Andrew Killian – 2nd Place in Best Presentation in Graduate Category in AIAA Region III Student Conference, Dayton, OH. [News Detail | AIAA](#)
- 2022: Jielong Cai – DCASS 2nd Place in Art-In-Science Competition https://www.aiaa-daycin.org/DCASS/docs/DCASS2023_AIS_Flier.pdf
- 2018: Asa Palmer - DESS Best Presentation in Undergraduate Student Presentation Competition
- 2018: Daniel Curry - Dayton-Cincinnati Aerospace Sciences Symposium Best Presentation Award
- 2018: Asa Palmer - Ohio Space Grant Consortium Best STEM Poster Award

PROFESSIONAL DEVELOPMENT

Certification Courses

1. Design and Simulation of Propeller Aerodynamics in Matlab, Udemy, May 22, 2020
2. Creating Apps in Matlab with App Designer, Udemy, May 18, 2020
3. Matlab Simulink Bible, Udemy, May 8, 2020
4. Control Systems: From Mathematical Modelling to PID Control, Udemy, May 11, 2020
5. Machine Learning for Data Science using MATLAB, Udemy, July 15, 2021
6. How to Use AI for Upskilling and Lifelong Learning, LinkedIn Learning, Aug 18, 2025
7. Beyond the Basics: Designing AI Enhanced Learning Experiences, LinkedIn Learning, Sept 18, 2025
8. Course: Nano Tips for Using Chat GPT to 10x Your Productivity at Work with Gianluca Mauro, LinkedIn Learning, Oct 7, 2025
9. Nano Tips for Future-Proofing Your Career with AI with Piers Linney, LinkedIn Learning, Oct 7, 2025

Teaching Workshops

1. KEEN EM Observable Behaviors Pilot Workshop, Milwaukee School of Engineering (MSOE), June 1st – June 3rd, 2025.
2. Developing Crescendo Workshops, Kern Family Foundation, Waukesha, WI, April 7th – 9th, 2025
3. Crescendo Workshop Showcase, Kern Family Foundation, Waukesha, WI, Aug 7th – 9th, 2025